

API Security Risk Analysis Report

Future Interns – Cyber Security Internship Task 3

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Date: May 2026

1. Introduction

Modern SaaS applications rely heavily on APIs for communication between services, applications, dashboards, and integrations. APIs expose critical business functionality and data, making API security an essential component of modern cybersecurity practices.

2. Objective

- Analyze public/demo APIs
- Assess authentication and access control mechanisms
- Inspect API responses and HTTP headers
- Identify security risks and weaknesses
- Recommend remediation strategies

3. Scope & Ethics

- Only public/demo APIs were tested
- Testing was limited to read-only requests
- No exploitation or malicious activity was performed

4. APIs Tested

ReqRes API

Authentication and security header analysis.

JSONPlaceholder API

Public endpoint and data exposure analysis.

5. Tools Used

- Postman
- Google Chrome
- Browser DevTools
- GitHub

6. Methodology

1. Reviewed API documentation
2. Tested endpoints using Postman
3. Inspected headers and responses
4. Documented findings and recommendations

7. Findings Summary

Finding	Severity
Authentication Enforcement Present	Informational
Public Endpoint Accessibility	Medium
Excessive Data Exposure	Medium
Missing Rate Limiting	High

8. Conclusion

This API Security Risk Analysis demonstrated common API security observations including authentication enforcement, public endpoint accessibility, excessive data exposure, and missing rate limiting controls. Regular API security assessments improve overall SaaS security posture and reduce organizational risk.